The World Leader Effect in Animal Welfare Policy

Categorising evidence of diffusion mechanisms in primary animal welfare legislation in the Danish, Austrian and German parliaments, 2009–2019

Student ID: 22001041834 The London Interdisciplinary School

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Abbreviations

Acronym	Meaning
FAW	Farmed Animal Welfare
EU	European Union

Country codes (ISO 3166)

Code	Country
DK	Denmark
AT	Austria
DE	Germany
NL	Netherlands
SE	Sweden
CH	Switzerland
UK	United Kingdom

In this study, 'A ◄ B' indicates country A is citing country B in its legislative process.

Abstract This pilot study asks how three Northern European countries make reference to each other and world leaders in their primary legislation[1] relating to farmed animal welfare between 2009 and 2019. Using the ParlLawSpeech V2 parliamentary corpus (Schwalbach et al., 2025), 453 documents, consisting of bills and laws, were assembled from the legislature of Germany, Austria, and Denmark. A regular-expression citation extraction pipeline produced 355 foreign country reference contexts, of which 62 formed the analytical sample after AI-assisted classification using the taxonomy of De Ruiter and Schalk (2017). Germany emerges as a discursive leader among follower countries, receiving 22 within-trio analytical citations, 20 of which come from Denmark. Qualitative analysis reveals that citations are mostly informational across all country pairs. This indicates that the operative diffusion mechanism for farmed animal welfare in the EU is learning rather than emulation. This is despite Denmark adopting most welfare reforms at the same time as or before Germany. Discursive attention that indicates learning under De Ruiter and Schalk's taxonomy may therefore also or instead be tracking competitive economic dependency. Future research should further this programme of improving the measures by which the diffusion mechanism of animal welfare policy can be unpicked as existing measures are still confounding.

1 Background

Over the last few decades, consumer concern regarding the implications of industrial animal agriculture has risen (Broom, 2017, p. 18; European Court of Auditors, 2018, p. 3). The EU moved quickly in the late 1990s and early 2000s to introduce minimum welfare standards (Broom, 2017, p. 24), and recognised animals as 'sentient beings' in the 2009 Treaty of Lisbon (European Court of Auditors, 2018, p. 3; Broom, 2017, p. 26). Supranational EU legislation has since stalled. Dairy cows, farmed fish, and rabbits are unprotected by specific EU directives (Broom, 2017, p. 54). European Citizens' Initiatives for a 'Fur Free Europe' and to 'End the Cage Age' both succeeded in gaining the requisite signatures but the European Commission is yet to bring forward a legislative proposal on either (European Commission, 2021, 2023).

Even the least progressive EU states remain comparatively advanced in the global context because of the EU setting a floor for animal welfare standards. However, EU minimums now often lag behind the progressive legislation of some of its member states and there is considerable variation between leader countries and laggards even within the EU (Vogeler, 2017). Five out of the six highest rated countries worldwide (ranked according to the World Animal Protection Index (API), each graded with a 'B' overall) are current or former EU member states (World Animal Protection, 2020). These are Austria, Denmark, Sweden, the Netherlands and the UK. The sixth, not an EU member, is Switzerland. It is the only country to have received the highest 'A' grade from the API, in 2014[2] (World Animal Protection, 2020). These six countries are this study's World Leaders.

2 Literature Review

Policy diffusion is 'the process whereby policy choices in one unit are influenced by policy choices in other units' (Maggetti & Gilardi, 2016, p. 87). There is consensus in the literature that diffusion is a product of interdependence. However, this interdependence can take different forms (Maggetti & Gilardi, 2016, p. 87). The literature refers to these forms as 'mechanisms', which in over a thousand articles (Shipan & Volden, 2012, p. 1) have been grouped into four broad categories: learning, emulation, competition and coercion (Shipan

[2] Switzerland was demoted to a 'B' grade in 2020, the same rank as this study's other World Leaders in animal welfare.

& Volden, 2008, p. 840), which for clarity are referred collectively to in this study as the LECC model.

Learning is defined as where 'the policies in one unit are influenced by the consequences of similar policies in other units' (Maggetti & Gilardi, 2016, p. 4). Emulation by contrast is not related to the objective consequences of a policy and instead its symbolic and socially-constructed characteristics are crucial (Maggetti & Gilardi, 2016, p. 5). Competition is the reaction between units in an attempt to attract and retain resources (Maggetti & Gilardi, 2016, p. 5). An imagined example of each might be: 'Swedish policy has proven effective in reducing sow mortality' (learning) versus 'we should be more like Sweden' (emulation) versus 'we need to mimic Sweden before Swedish producers undercut us' (competition). Although coercion is not a mechanism this study examines, it is defined as when threats are made to subordinate units if they do not comply with policy changes suggested by the senior unit, e.g. 'The EU will fine us if we do not adopt animal welfare standards'.

Graham, Shipan and Volden (2013) conducted a systematic review documenting how diffusion research has itself 'diffused' across the social sciences by means of the shared conceptual vocabulary of these four mechanisms. Mallinson (2021) extended this in a meta-review of policy diffusion studies in American states, estimating effect sizes of the most commonly used diffusion variables and identifying biases in published research. The cumulative picture is that diffusion effects are real and robust, but their magnitude and operative mechanisms vary substantially across policy types and political contexts. Maggetti and Gilardi (2016) provide a meta-analysis of 114 studies demonstrating that the same indicators are used to measure different diffusion mechanisms and different indicators are used to measure the same diffusion mechanism (Maggetti & Gilardi, 2016, p. 18).

There are multiple criticisms of the LECC model in the literature which have resulted in alternative proposals to redress its perceived shortcomings. The first, the underspecification of these four terms, has led to the development of new definitions to be applied in a more strictly deductive fashion. Blatter et al. (2021) address bias, and reformulate the mechanisms into interest-driven, rights-driven, ideology-driven, and recognition-driven pathways. Gilardi and Wasserfallen (2019) are concerned by a failure of omission by the model. They argue that the LECC model underspecifies the political dimensions of the process, and treats the selective borrowing of foreign policy as a well-calculated choice by legislators trying to garner legitimacy for their proposals. Genovese et al. (2016) corroborate this and show that such an assumption would be a systematic error. Governments are rarely presented with a single diffusing policy in isolation. There are frequently alternative policies to consider, but a country must also consider their dependency on foreign resources and will choose a substitute policy by evaluating the more limited options available under these constraints.

Linos argues instead that diffusion and domestic democracy are mutually reinforcing (Linos, 2013, p. 2). Politicians emulate the policy choices of familiar, wealthy and culturally proximate countries to signal to 'uninformed' voters that their proposals are mainstream, competently designed and vetted (Linos, 2013, p. 3).

Despite these legitimate criticisms and modifications of the most widespread definition framework (LECC), this study chooses to persist with using the LECC definitions because in so doing 'we open our work up to more natural comparisons to similar studies elsewhere' (Graham et al., 2013, p. 700). Placing this capstone in a lineage of studies that share these familiar terms offers advantages. It can build on existing insights as well as enabling more straightforward verification and reproduction of results. To mitigate the pitfalls of the LECC framework, this study requires explicit and systematic reflection between how concepts are being conceptualised and how they will be operationalised. This will be laid out in the methodology. The challenge of the LECC framework is in how to consistently test its mechanisms in different contexts. The background definitions of these concepts, before they are systematised, are uncontentious.

Policy diffusion operates across multiple governance levels. Countries can diffuse policies between one another in a peer-to-peer horizontal process (De Ruiter & Schalk, 2017, p. 2). Particularly pertinent in the European Union context is vertical influence. 'Bottom-up' diffusion occurs when individual member states influence the EU parliamentary process. 'Top-down' diffusion involves EU mandates reshaping member states' behaviour (Shipan & Volden, 2006, p. 825). It is in top-down instances that coercion is most commonly observed. Now that EU-level farmed animal welfare legislation has stalled, the operative diffusion mechanism has therefore shifted from top-down to peer-to-peer.

Policy diffusion literature leverages the methods of analysis of many different social science disciplines. All the mechanisms described above have social, political and economic dimensions to them. The significant influence of these contextual factors makes it difficult to develop accurate, predictable theories of policy diffusion at scale. It is therefore essential to stratify the analysis as much as possible by limiting the research parameters to well-established geographic or federal blocs e.g. The USA or the EU. The focus is less on the specific policy or category under investigation and more about developing a better theory of the *mechanics* of policy diffusion between similar countries or states. Interestingly, although this recognition is well-aligned to the axioms of complexity science and its pursuit of meta-parsimonious explanations, the author did not find any literature on policy diffusion or animal welfare policy conducted through that prism.

The terms 'leader' and 'follower' are widely used in policy diffusion literature. In different instances, a policy leader is defined as: the first mover in an adoption sequence; the most cited or emulated country in the debates of adopting states (discursive leadership); the state whose legislative turn of phrase is most extensively reused; or the state exerting the greatest causal influence on the probability of adoption elsewhere. This study focuses on the second definition of a policy 'leader' as the country most frequently cited when legislators in other jurisdictions debate animal welfare legislation[3].

Members of Parliament operate in an environment of 'attention scarcity' (Jones & Baumgartner, 2005, as cited in De Ruiter & Schalk, 2017, p. 1), meaning they must find ways to lower 'information collection costs', in other words reduce complexity when identifying the most critical issues to address (De Ruiter & Schalk, 2017, p. 1). One key strategy is referencing the policy practices of other countries during plenary debates. De Ruiter and Schalk (2017, p. 4) demonstrate that MPs use new information about the policy practice in another country in parliamentary debate as it becomes available and citations could be a lever of legitimation. Linos (2013, p. 3) provides a theoretical basis for this, hypothesising that this is to signal a proposal is mainstream and internationally vetted, and therefore normatively acceptable to peers (and voters). There are other sources of policy that could be used as exemplars (e.g. think-tanks) but they are perhaps perceived as less rhetorically effective in debate.

De Ruiter and Schalk (2017) inductively develop a taxonomy of foreign country references in Dutch plenary speeches that distinguishes between informational citations (where a foreign policy is described to inform deliberation) and legitimating citations (where the foreign case is invoked to validate a domestic proposal). A third type, cautionary citations, cite a foreign case as a warning or negative example. Note that citation frequency analysis captures salience in the debate, not necessarily cause and effect. A country can be cited frequently as a cautionary tale. It can be cited because it is culturally proximate and well-known rather than because its specific policy design has been substantively vetted and appropriated (i.e. whether the policy is diffusing via emulation or learning).

Policy diffusion does not solely occur at the final stage of policy adoption. A significant but often neglected aspect of the diffusion process is the issue-definition stage (Gilardi et al., 2021, p. 21). Before policies are adopted or placed on an agenda, issues must be identified and defined through specific policy frames (Gilardi et al., 2021, p. 22). Studying diffusion solely through the lens of successful adoptions leads to a 'pro-innovation bias'. Policies pass through various framing stages and adoptions themselves are rare (Gilardi et al., 2021, pp. 22–23). This study, therefore, includes proposed bills that did not pass alongside enacted laws.

This research focuses specifically on laws and bills concerning farmed animal welfare in Denmark, Germany, and Austria between 2009 and 2019. This ten-year period is the overlap between the available data from ParlLawSpeech for these three countries.

From this literature review the following sub-questions have emerged:

1. Which foreign countries appear most frequently in the animal welfare legislation of each citing country, and does this pattern differ across the trio?
2. Do those citations function primarily as evidence-gathering (informational), social proof (legitimating), or competitive awareness (cautionary)?
3. Does the direction of discursive attention align with the sequence in which farmed animal welfare legislation was adopted?

[3]This is for reasons expanded on in the methodology, but briefly because it can be measured directly from parliamentary records using the citation analysis methods described in the methodology section.

3 Methodology

This study uses a convergent parallel mixed-methods design (Creswell & Plano Clark, 2011), in which a quantitative strand (citation frequency) and a qualitative strand (citation classification) are run in parallel and integrated in a final synthesis table.

ParlLawSpeech is a multilingual parliamentary corpus covering legislative documents from multiple European countries (Rauh et al., 2023). For this study, the relevant components are the bills and enacted laws from Germany, Austria, and Denmark. The corpus provides documents in their original languages along with metadata including adoption date, initiation date and an identifier for each legislative item. This last links all documents belonging to the same legislative process. It can be used to recover laws whose keyword coverage in the enacted text is lower than in the originating bill.

Foreign country citations were extracted using regular expression matching on the full text of each document in the corpus. Country terms were defined in a `COUNTRY_NAMES` dictionary covering each target country's name in German, Danish, and English, with word-boundary anchors (`\b`) to prevent false matches. This prevents compound words such as *deutschnationale*, [German nationalist] from registering as a Germany reference. The target countries were Germany, Austria, Denmark, Sweden, Switzerland, the Netherlands, and the United Kingdom. Sweden and Switzerland were included as reference countries known from the adoption sequencing literature (Wallenbeck et al., 2024) to be relevant to the regional welfare diffusion context.

For each regex match, a context window of ± 600 characters around the match was extracted to enable subsequent classification. Each match generated one row in the citation dataset: one document can contribute multiple rows if the target country is mentioned more than once.

Each of the 355 extracted citation contexts was classified using the Claude Haiku 4.5 language model via the Anthropic API. The classification scheme was based on De Ruiter and Schalk's (2017) taxonomy for plenary speeches, in this case deployed on bills and laws and extended to include three exclusion categories.

Analytical categories (can co-occur).

- *Informational:* the foreign case is cited to provide factual information about a policy elsewhere; function is to inform deliberation.
- *Legitimating:* the foreign case is cited to validate or authorise a domestic proposal; function is to signal the proposal is mainstream and peer-vetted.

- *Cautionary:* the foreign case is cited as a warning or negative example.

Exclusion categories (mutually exclusive).

- *False positive:* the country name appears but not as a reference to that country as a political actor or policy exemplar (e.g. language references such as *deutsche Übersetzung*, proper nouns, geographic descriptions).
- *Out of scope:* the country is cited as a policy actor, but the document concerns non-farmed animal welfare (companion animals, zoo animals, wildlife).
- *EU implementation:* the document is transposing or ratifying EU legislation; the country is mentioned as a co-signatory, not as a voluntary peer whose policy is being studied.

Before examining the AI output, the author independently classified 15 randomly sampled contexts and recorded labels. The sample was locked.

4 Results

The filtering pipeline retained 453 documents in total: 204 laws and 249 bills. Denmark contributes the largest share of both categories (laws: 56%, bills: 51%). This reflects the larger size of the Danish component of ParlLawSpeech for this period. Germany and Austria each contribute roughly equally to the remainder.

Regular expression extraction produced 355 foreign country citation contexts across the 453 documents. Denmark generated the largest share of raw citations (213, 60%), followed by Austria (91, 26%) and Germany (51, 14%). This asymmetry partly reflects Denmark's larger corpus share and the denser text of Danish bills. Denmark produces roughly twice as many foreign country mentions per document as Germany (0.89 versus 0.43 citations per document). Austria's per-document rate (0.96) is slightly higher, though Austria's smaller corpus means its absolute count remains far behind Denmark's.

The most-cited country by raw frequency is Switzerland (94 citations), followed by Germany (93) and Sweden (66). Austria receives 22 raw citations; Denmark receives 10; the Netherlands 33; the UK 37.

Switzerland's high raw count reflects EU implementation documents. Switzerland is frequently named as a co-signatory or comparative case in EU documents. This is consistent with Switzerland's institutional position vis-à-vis the EU. Since 1995, Switzerland has pursued a policy of autonomous regulatory alignment with EU veterinary and food safety law, formalised from 2002 in the bilateral agreement on trade in agricultural products and extended in 2009 to a Common Veterinary Area covering animal health and welfare standards.

Of the 355 citation contexts, 293 (82.5%) were excluded:

- *EU implementation:* 87 (24.5%)
- *False positive:* 102 (28.7%)
- *Out of scope:* 104 (29.3%)

The remaining 62 contexts (17.5%) formed the analytical sample. The high exclusion rate is expected. Country names appear frequently in legislative documents for incidental reasons, such as a part of EU regulatory language or in legislation about zoo or companion animal welfare.

The distribution of labels across the 62 analytical citations is:

- *Informational:* 58
- *Cautionary:* 6
- *Legitimating:* 2

Analytical categories can co-occur (and do, in this case). Therefore there are more than 62 citations.

Across all analytical citations in the corpus, the dominant function of foreign country references is informational. Using De Ruiter and Schalk's (2017) framework it appears that citations of other countries are primarily to describe what those countries have already done.

Table 1: Citation mechanisms by country pair

Citing ◀ Cited	N	Inf.%	Leg.%	Caut.%	Mechanism
DK ◀ DE	20	90	0	20	Learning
DE ◀ DK	3	100	0	0	Learning
AT ◀ DE	2	100	50	0	Learning + Emulation
DE ◀ AT	2	50	50	50	Learning + Emulation
AT ◀ DK	1	100	0	0	Learning
DK ◀ AT	0	—	—	—	—

Germany receives 22 analytical citations, 20 from Denmark and 2 from Austria. Denmark receives 4, 3 from Germany and 1 from Austria. Austria receives 2, both from Germany.

Denmark continues to cite higher than Austria and Germany when extended to World Leaders. The full integration table records DK ◀ SE (11 analytical citations) as the second-strongest citation relationship in the dataset. Denmark's outward orientation is not exclusively directed at Germany. Sweden's prominence in Danish legislative discourse is consistent with its role as the earliest adopter of pig welfare reforms in the region.

Across all analytical citations ($N = 62$), Denmark accounts for 40 of them (65%). Germany 14, and Austria 8. Denmark is the most prolific citing country and the recipient of the fewest within-trio citations. Germany is the least prolific citing country and the most-cited.

The raw Cohen's Kappa, comparing the author's labels to the AI's classifications, was $\kappa = 0.281$. This suggests fair agreement on the Landis and Koch (1977) scale.

This figure is misleading in a specific and important way. Eleven of the fifteen validation cases concerned the boundary between *false positive* and *out of scope*. These categories both lead to exclusion from the retained sample. The author classified several cases as *out of scope* that the AI classified as *false positive*, or vice versa. This boundary is admittedly ambiguous. This is because a document about zoo animal welfare that mentions Germany could be argued either way, depending on whether the country is read as a genuine policy exemplar or merely an incidental mention.

When the three exclusion categories are collapsed into a single 'excluded' label, agreement on the analytically consequential include-versus-exclude decision rises to 14 out of 15 cases (93.3%). The Cohen's Kappa on this binary include/exclude decision is $\kappa = 0.762$[4]. The author classified one case as *unclear* (which would have included it in the analytical sample) while the AI classified it as *false positive*, which excluded it.

The two cases that both author and AI agreed should be included in the analytical sample were independently assigned the 'informational' label. One case where the author assigned 'cautionary' in addition to informational was classified by the AI as informational only.

These results were checked against temporal adoption sequencing data from Wallenbeck et al. (2024), who documented the year in which each country adopted specific pig welfare measures relative to Sweden. For the study period, the relevant in-corpus measures are gestation housing (Austria in 2012, Germany in Denmark in 2013). This was the only welfare adoption event occurring for all three study countries within the 2009–2019 window at a point captured by the corpus.

4.1 Diffusion Mechanisms

The dominant mechanism across the corpus is learning according to the deductive framework. 93.5% of analytical citations are informational, whereas legitimating citations make up 3.2% (2/62). The most extensively cited relationship (Denmark to Germany, $N = 20$) is 90% informational, 0% legitimating, and 10% cautionary.

All six cautionary classifications identified overall appear in three specific citation relationships: DK ◀ DE, DE ◀ AT, and DE ◀ NL. The DK ◀ DE cautionary cases concern disease outbreak risk. Danish legislation cites Germany's 2011–2012 dioxin crisis and historical swine fever outbreaks as exemplars of what can happen when food safety oversight fails. The DE ◀ AT and DE ◀ NL cautionary cases concern fur farming. A 2017 German bill cites Austria, the Netherlands and others as countries that have already banned mink farming, positioning Germany as a laggard relative to its European peers.

Switzerland generates the highest number of raw citations across the corpus (94) but only 6 analytical citations after exclusion. Germany also cites Switzerland ($N = 3$) and the Netherlands ($N = 3$) in its analytical sample, primarily in the fur farming context. A 2017 German bill on mink welfare (mentioned in Vignette 3, below) compares German minimum space requirements unfavourably with Swiss, Austrian, and Dutch provisions. This document is responsible for the entirety of the DE ◀ AT and DE ◀ NL cautionary classifications.

4.2 Illustrative Vignettes

Vignette 1: Cautionary, DK ◀ DE (2012, disease control)

A 2012 Danish bill on veterinary food safety controls invokes Germany's dioxin contamination crisis to argue for more robust domestic oversight. The context reads: *'det danske kontrolsystem hidtil vist sig så effektivt, at vi har undået kemiske forureninger'*. The Danish control system has so far proven so effective that Denmark has avoided the chemical contamination incidents [of the kind seen in Germany]. Germany is cited as a warning. Denmark defines its own system's quality by contrast with Germany's failure. This is a competitive-awareness citation with a strong cautionary function. The AI classification (confidence: high) characterises this as a case where Germany's experience is invoked 'to warn against potential economic consequences Denmark might face if similar contamination occurs, establishing a cautionary comparative reference.'

Vignette 2: Informational, DK ◀ DE (2018, veterinary medicine)

A 2018 Danish veterinary medicine bill surveys how comparable EU countries handle the dispensing of measured quantities of veterinary drugs. The relevant passage names *'Belgien, Finland, Tyskland og Holland'*, Belgium, Finland, Germany, and the Netherlands, as countries where such dispensing schemes exist in some form. It also notes that in all these countries (as in Denmark) vets can already administer precise doses during treatment. Germany appears here as one data point in a comparative audit, cited descriptively and without evaluative language. The AI classification (confidence: high) characterises this as 'providing factual information about how other member states handle medication portioning without invoking Germany as a model for legitimation or warning.'

Vignette 3: Cautionary, DE ◀ AT (2017, fur farming)

A 2017 German bill on fur animal welfare cites Austria, the Netherlands, the United Kingdom, and several other European countries as jurisdictions that have already enacted fur farming bans. The context notes that these countries have moved ahead on animal protection standards. Germany however, despite recommendations to the contrary from mammalogy experts, still requires only one square metre of space per mink. Austrian and Swiss requirements are many times larger. Austria is cited as evidence that Germany is falling behind. The cautionary function is self-directed. The AI classification (confidence: high) reads this as functioning 'both to inform about existing policy elsewhere and to warn that Germany is lagging behind peer nations in animal protection standards.'

5 Discussion

The results are consistent with Mallinson's (2020) finding that ideological learning becomes more prominent in later stages of 'the [policy] life course', displacing the regional proximity effects dominant in early stages.

Germany receives the most within-trio citations in farmed animal welfare legislation, driven overwhelmingly by Denmark. On the standard operational definition used in this study, Germany is the discursive leader.

However, Germany does consistently not lead Denmark in welfare adoption. Denmark adopted tail docking and manipulable material restrictions earlier than Germany and reached

[4] $0.762 = (0.933 - P_e)/(1 - P_e) \therefore P_e \approx 0.719$

gestation housing reform at the same time. There is no empirical basis for treating Denmark as a discursive follower of a policy pioneer, yet Denmark cites Germany more than Germany cites Denmark, and more than either Austria or Germany cite each other.

One plausible explanation is competitive monitoring. Germany is Denmark's largest neighbour, a major pork-producing economy, and a competitor in European export markets (Denver et al., 2021). Danish legislators have incentives to track German welfare legislation to know when Germany will be subject to the same costs that Danish producers already bear. If Danish pigs are raised under stricter welfare conditions than German pigs, Danish pork is more expensive to produce (Vogeler, 2017). The moment Germany adopts equivalent standards, that cost disadvantage may narrow. Denmark's parliament has an interest in knowing when Germany will reach that point.

This pattern is consistent with recent comparative work on regulatory competition in farm animal welfare elsewhere. Zöllmer (2025), examining the diffusion of welfare standards across US states, argues against the assumption that competitive exposure straightforwardly suppresses unilateral welfare improvement. The governance mode through which a standard is implemented determines whether competitive pressure produces resistance or accommodation (*cf.* Vogel's (1995) account of 'races to the top' in environmental regulation).

Legitimating citations across the corpus are extremely rare, while De Ruiter and Schalk (2017) found sufficient legitimating citations to inductively invent a category for them in the Dutch parliamentary context. The author offers several possible explanations. The sample is made up by bills and laws, not speeches. Legitimating rhetoric may be more characteristic of oral parliamentary debate than of written legislative texts, where the primary function is regulatory specification. A study of parliamentary speeches may reveal more legitimating citation. Another is that the countries in this study are still relatively high-welfare both in the European and global contexts. Legitimating citations may be more characteristic of laggard countries catching up. A study that included lower-welfare EU member states as citing countries would enable a direct test of this life-course interpretation.

5.1 Limitations

Several limitations should be noted. First, the corpus excludes parliamentary speeches for the primary analysis, meaning that the full discursive record is not examined. Second, the Kappa of 0.281 reflects genuine disagreement between the author and the AI on exclusion subcategory boundaries, even if the include/exclude agreement was 93.3%. The AI may have overclassified or underclassified some analytical cases in ways that affected the mechanism distribution. Third, the small N in some country pairs (AT ◀ DK: 1, AT ◀ DE: 2, DE ◀ AT: 2) means that mechanism classifications for those pairs are indicative rather than robust. Fourth, some DK ◀ SE citations in the analytical sample appeared on closer inspection to concern non-welfare topics (aviation regulation, excise duties) present in multi-topic bills that passed the keyword filter; this suggests the broad-tier keyword filter includes some documents with only marginal welfare relevance, and that the out-of-scope classifier did not catch all instances.

Fifth, the study's chosen taxonomy was developed inductively by De Ruiter and Schalk (2017) from Dutch parliamentary debate on education and environmental policy. The categories of informational, legitimating, and cautionary citation may carry different rhetorical weight in German, Danish and Austrian legislative systems, and in the specific domain of farm animal welfare. The near-absence of legitimating citation in this corpus could also reflect a difference between nations in how foreign references function. This study cannot disentangle this. A validation of the taxonomy against a corpus from a country or policy domain where De Ruiter and Schalk's original categories were directly observed would help establish how portable the taxonomy is.

Future research should extend the analysis to parliamentary speeches, which would provide a larger and more rhetoric-rich corpus. Including lower-welfare EU member states as citing countries would enable comparison of legitimating citation rates across welfare levels. It would provide a more direct test of the life-course interpretation offered in the discussion. A systematic review of the legislative language borrowed across jurisdictions would complement the citation frequency approach used here and triangulate a clearer measure of policy transfer.

6 Conclusion

Using the ParlLawSpeech corpus, 453 legislative documents were assembled and 355 foreign country citation contexts were extracted and classified using an AI-assisted taxonomy derived from De Ruiter and Schalk (2017). Germany emerges as the discursive leader, receiving 22 within-trio analytical citations compared to 4 for Denmark and 2 for Austria. The mechanism appears to be predominantly learning, as informational citation accounts for 93.5% of analytical cases, with legitimating citations very rare. The cautionary cases concern two distinct phenomena: disease outbreak risk (DK ◀ DE) and laggard positioning relative to peers in fur farming welfare standards (DE ◀ AT, DE ◀ NL).

However, the study demonstrates an asymmetry between adoption sequencing and what discursive attention would indicate under De Ruiter and Schalk's taxonomy. Denmark adopted several welfare measures at the same time as or before Germany, yet Denmark monitors Germany's legislative debates far more intensively than Germany monitors Denmark's. This suggests that discursive leadership, as measured by citation frequency, may reflect competitive economic dependency, and not learning. Germany may function as Denmark's discursive reference point due to this economic dependency rather than because German welfare policy is more advanced.

Countries cite each other for different reasons at different stages of the diffusion lifecycle. A country that cites frequently may be learning, emulating, or watching a competitor.

Disambiguating these mechanisms is essential for connecting the legislative process to how policy diffusion occurs.

This work is 5,905 words in length from the title to the end of the reference list.

Verification data, code, supplementary information, and the Capstone product are available at github.com/thisisgoingtobechanged/l6_capstone.

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